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Advanced deep learning architectures for enhanced mammography classification: a comparative study of CNNs and ViT

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Abstract

Breast cancer is a leading cause of mortality among women worldwide, and early detection via mammography is critical for improving patient outcomes. In this study, we conduct a comprehensive comparative analysis of 15 state-of-the-art deep learning models including both general-purpose (e.g., ResNet50, ConvNeXt, ViT) and mammogram-specific architectures (e.g., FCCS-Net, ViT-Mammo, GLAM-Net) for breast cancer classification using mammographic images. Leveraging four publicly available datasets, we evaluate all models using a unified and reproducible pipeline under standardized training protocols. Beyond traditional performance metrics, our multi-dimensional assessment includes interpretability via Grad-CAM and attention maps, calibration reliability, inference time, model complexity, and deployment feasibility. Our findings highlight the superior diagnostic accuracy and visual interpretability of mammogram-specific models, particularly FCCS-Net and ViT-Mammo, with ViT-Mammo achieving an AUC of 0.961. Lightweight architectures such as EfficientNetB0 and DenseNet121 demonstrate strong potential for edge deployment due to their efficiency and competitive accuracy. The study further explores model robustness across datasets, calibration reliability, and real-time deployment constraints, offering actionable insights for clinical integration of AI-driven diagnostic tools. This work provides a valuable benchmarking framework and paves the way for the development of interpretable, efficient, and clinically viable AI systems for breast cancer screening.

Keywords Breast cancer detection, Mammography classification, Deep learning, Convolutional neural Networks (CNN), Vision transformer (ViT), Computer-aided diagnosis (CAD) systems

1 Introduction

Breast cancer is the most commonly diagnosed malignancy among women worldwide and remains a leading cause of cancer-related deaths [1]. Timely and precise detection through screening is essential for enhancing patient outcomes and survival rates. Among available imaging techniques, mammography remains the gold standard for early breast cancer diagnosis due to its non-invasive nature and cost-effectiveness. However,

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interpreting mammographic images (Fig. 1) remains a complex and error-prone task, influenced by subtle visual cues, varying tissue densities, and inter-observer variability among radiologists [2]. These challenges underscore the critical need for diagnostic tools that offer greater objectivity and reliability.

In recent years, deep learning (DL) has emerged as a transformative paradigm in medical image analysis, offering significant promise for automating disease detection and classification tasks [3, 4]. Early convolutional neural network (CNN) models such as VGG [5] and ResNet [6] demonstrated encouraging results in identifying abnormalities in mammograms. However, these conventional CNNs often suffer from limitations in generalization, model calibration, and interpretability—factors that are essential for clinical trust and adoption [7].

The DL landscape has since evolved with the advent of state-of-the-art (SOTA) architectures, including Vision Transformers (ViTs) [8], Swin Transformers [9], ConvNeXt [10], and hybrid models such as CoAtNet [11]. These models have consistently outperformed earlier CNNs across computer vision benchmarks and are increasingly being explored for medical imaging applications [12, 13]. Furthermore, the development of mammogram-specific architectures such as MammoNet, GLAM-Net, ViT-Mammo, and FCCS-Net marks a critical shift toward domain-tailored AI systems. Yet, despite these advances, a rigorous, unified comparison of general-purpose and mammogram-specific models under clinically relevant conditions remains lacking.

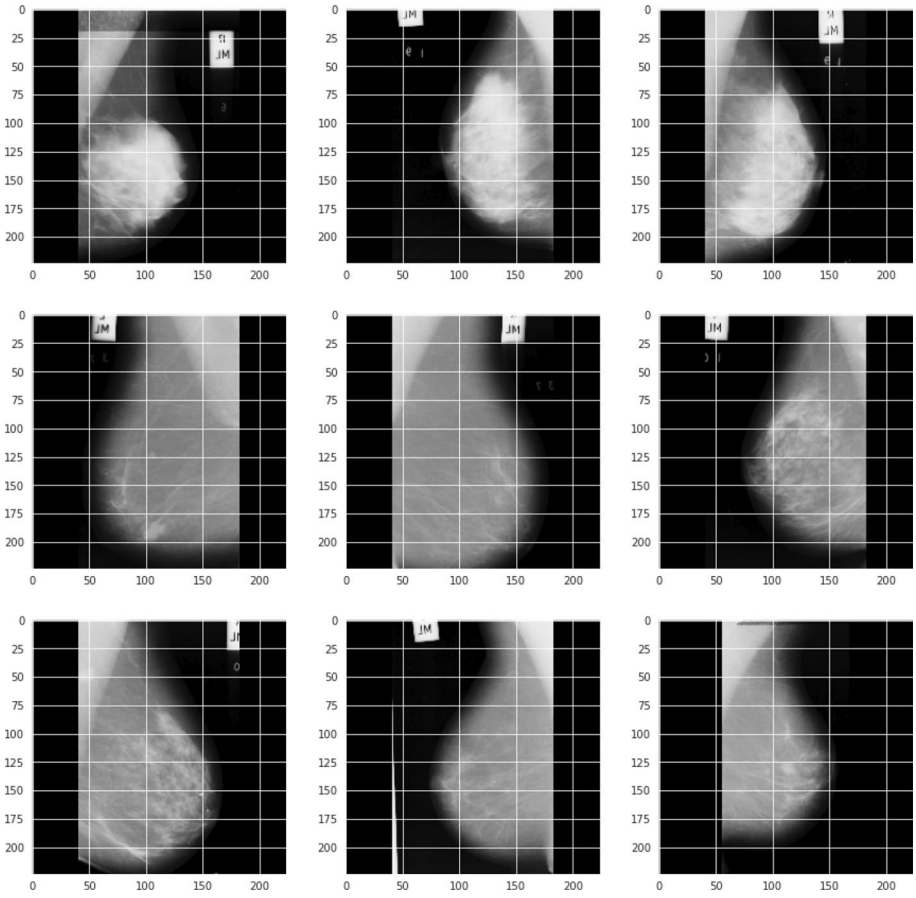


Fig. 1 Mammography images of various patients

This study addresses that gap by providing a comprehensive evaluation of recent deep learning models for breast cancer detection in mammography. We benchmark 15 SOTA architectures, including general-purpose models (e.g., ResNet50, ViT, EfficientNetB0) and mammogram-specific models (e.g., FCCS-Net, MammoNet, ViT-Mammo), using a unified experimental framework across four public mammography datasets. In doing so, we aim to assess not only model accuracy, but also interpretability, calibration reliability, and deployment feasibility—critical aspects for real-world clinical integration.

1.1 Key contributions

This study introduces several novel contributions, including:

- *Comprehensive evaluation:* We conduct an extensive comparison of 15 deep learning models, spanning both general-purpose and mammogram-specific architectures, for breast cancer detection using mammography.
- *Standardized training protocols:* All models are trained under identical hyperparameter settings and evaluated using consistent training, validation, and test splits across four public datasets (CBIS-DDSM, INbreast, MIAS, and VinDr-Mammo), ensuring reproducibility and fairness.
- *Multi-dataset benchmarking:* We perform cross-dataset evaluation to assess model robustness and generalizability, addressing a critical but often neglected factor in breast cancer AI research.
- *Interpretability analysis:* Using attention heatmaps and Grad-CAM visualizations, we assess each model's ability to localize lesions, contributing to the interpretability and trustworthiness of AI-assisted diagnosis.
- *Deployment feasibility:* We provide a computational trade-off study covering inference time and memory consumption on both CPU and GPU, highlighting models suitable for resource-constrained clinical settings.
- *Open benchmarking framework:* To facilitate future research, we release a reproducible and extensible benchmarking pipeline for automated breast cancer diagnosis using mammograms.

The remainder of this article is organized as follows: Sect. 2 reviews existing literature and traces the development of deep learning architectures in medical imaging. Section 3 details the datasets, preprocessing techniques, model architectures, and experimental design employed in this study. The results are presented in Sect. 4, with a comprehensive analysis provided in Sect. 5. Finally, Sect. 6 summarizes the key findings and outlines potential avenues for future research.

2 Related work

Breast cancer continues to be a major contributor to cancer-related deaths among women globally, highlighting the urgent need for precise and early diagnostic strategies. Recent advances in medical imaging and artificial intelligence (AI), particularly deep learning, have greatly enhanced breast cancer detection, outperforming traditional methods.

2.1 Traditional methods and their limitations

Mammography has long served as the primary modality for breast cancer screening. Despite its widespread adoption, it faces notable limitations—most prominently, its variable sensitivity in detecting cancer, particularly in women with dense breast tissue. Kerlikowske et al. [14] reported that in such cases, the sensitivity of mammography can decline to as low as 30%, impacting a significant portion of the screened population. In response to these challenges, advancements such as digital mammography and tomosynthesis have been introduced, offering improved image quality and mitigating the issue of overlapping tissues. Nevertheless, these newer modalities still exhibit a high rate of false positives, often resulting in unnecessary biopsies and increased patient anxiety, as highlighted by Berg et al. [15].

Rezaei and Zahra [16] provided a comprehensive review of image-based techniques for breast cancer detection, segmentation, and classification, examining both automatic and semi-automatic methods developed between 2016 and 2020. Their work discusses diagnostic methodologies, associated risk factors, and the evolution of imaging technologies. Additionally, Tahmooresi et al. [17] proposed a hybrid diagnostic framework that integrates multiple machine learning algorithms, including Support Vector Machines (SVM), Artificial Neural Networks (ANN), K-nearest neighbors (KNN), and Decision Trees (DT), demonstrating its efficacy in enhancing breast cancer detection. The study explores critical components of the breast cancer diagnostic pipeline, including image preprocessing, segmentation, feature extraction, and tumor classification. It also introduces several publicly available breast imaging datasets to support research and development in the field. In addition to outlining current methodologies, the study addresses prevailing challenges, practical limitations, and future research directions in breast imaging technologies. While many existing approaches rely on a single imaging modality, accurate diagnosis and personalized treatment often benefit from multi-modal imaging. However, integrating diverse imaging sources generates large volumes of data, necessitating scalable solutions such as cluster-based computing frameworks for efficient processing and analysis.

Elleuch et al. [18] proposed a novel possibilistic modeling framework for decision-making in breast cancer screening. Their method incorporates clustering-based feature aggregation, followed by the fusion of possibility distributions to enhance diagnostic performance. This approach was evaluated through two case studies focused on detecting microcalcifications and masses—both of which present considerable diagnostic challenges. Validation was performed using the CBIS-DDSM public dataset, with the model achieving impressive detection accuracies of 95.4% for masses and 99.4% for microcalcifications, along with specificities of 96% and 100%, and corresponding AUCs of 0.96 and 0.996. Further highlighting the momentum in this domain, Shah et al. [19] discussed how recent advancements in deep learning, computational infrastructure, and the increasing availability of annotated datasets have significantly fueled interest in AI-driven approaches for breast cancer detection.

Researchers have investigated a range of imaging modalities, including mammography, ultrasonography, MRI, and histopathology, for automated breast cancer diagnosis. This article reviews the advantages, limitations, and datasets associated with these imaging techniques. It also highlights recent AI-based frameworks, with a focus on mammography, the most commonly used screening method. The availability of annotated

mammography datasets plays a crucial role in training AI models, significantly influencing performance. This study provides valuable insights into AI techniques and imaging modalities, making it an important resource for AI-driven breast cancer screening. While advances in early screening have enhanced survival rates, image interpretation continues to be labor-intensive and prone to errors. AI, particularly Machine Learning (ML) and Deep Learning (DL), helps improve diagnosis by assisting in lesion localization and reducing the risk of misdiagnosis. This review examines the current state of mammography with ML and DL, highlighting recent breakthroughs in DL for mammogram analysis and outlining future research directions [20].

2.2 Emergence of deep learning in breast cancer detection

The integration of deep learning into breast cancer detection has shown significant promise in overcoming the limitations of traditional diagnostic methods. Convolutional neural networks (CNNs), in particular, have demonstrated exceptional capabilities in image classification and pattern recognition. Esteva et al. [21] were among the first to apply CNNs for skin cancer classification, setting a precedent that spurred similar advancements in breast cancer research. Houssein et al. [22] present a comprehensive review of machine learning and deep learning applications in breast cancer detection and classification, with a focus on multi-modal medical imaging. Their work examines methods for distinguishing between tumors, non-tumors, and dense masses across various imaging techniques. The review begins with an overview of traditional machine learning approaches and proceeds to an in-depth discussion of deep learning methods and architectures tailored for breast cancer diagnosis. It also encompasses a broad spectrum of imaging modalities, offering a comprehensive perspective on the field. By analyzing a wide range of research databases, the study highlights emerging trends, existing challenges, and future prospects in breast cancer detection and classification.

In [23], the authors propose a computer-aided diagnosis (CAD) system based on deep learning (DL) to support radiologists in classifying breast cancer lesions in mammograms. The study explores four experimental approaches to identify the most effective strategy. The first involves fine-tuning pre-trained deep convolutional neural networks (DCNNs). The second extracts deep features from these networks and employs support vector machines (SVMs) with different kernels for classification. In the third experiment, deep features are fused to enhance SVM classification accuracy. The fourth approach incorporates principal component analysis (PCA) to reduce feature dimensionality and computational load. Using the CBIS-DDSM and MIAS datasets, the experiments demonstrate that deep feature fusion yields the highest classification accuracy, whereas PCA effectively reduces execution time without improving accuracy.

McKinney et al. [24] conducted a landmark study in which they developed a deep learning model trained on over 76,000 mammograms. Their model surpassed radiologists in breast cancer detection accuracy, significantly reducing both false positives and false negatives. This work marked a pivotal step forward, illustrating that AI systems can not only match but potentially outperform human experts in specific diagnostic tasks. Building on this progress, Tsochatzidis et al. [25] introduced an innovative approach that enhances breast cancer diagnosis by incorporating mass segmentation data into a convolutional neural network (CNN). Their method modifies the convolutional layers to process both the original mammographic image and its corresponding segmentation

map. A novel loss function was also proposed, augmenting the standard cross-entropy with an additional term that directs the model's focus toward the mass region while penalizing irrelevant activations. Evaluated on the DDSM-400 and CBIS-DDSM datasets, the approach achieved AUC scores of 0.898 and 0.862 using ground-truth segmentation, and up to 0.880 and 0.860 with U-Net-generated segmentation maps. These results confirm that integrating segmentation information can significantly boost CNN performance in mammographic mass classification.

In [26], the authors present a comprehensive review of computer-aided techniques for mammographic breast cancer detection, focusing on the segmentation and classification of microcalcifications from 1970 to 2023. Adhering to PRISMA guidelines, the review explores the progression of methodologies in image processing, machine learning, and deep learning—tracing the shift from early AI models to sophisticated deep learning architectures. The study highlights the development of automated diagnostic systems, evaluates deep learning approaches utilizing self-extracted features, and underscores the importance of large-scale datasets in driving future research. In line with the United Nations' sustainability goals, it also emphasizes the transformative potential of AI in promoting early and accurate breast cancer detection [27].

2.3 Advanced deep learning techniques

Recent advancements have led to the development of more sophisticated deep learning architectures. For instance, Wu et al. [28] proposed an innovative deep learning model that integrates both mammograms and clinical data, resulting in enhanced diagnostic accuracy. This multimodal approach is particularly valuable as it mirrors real-world clinical practice, where radiologists consider both imaging and patient history when making diagnostic decisions.

A significant advancement in breast cancer detection is the application of transfer learning, where models pre-trained on large datasets are fine-tuned for specific medical tasks. Ragab et al. [29] demonstrated that transfer learning can substantially reduce the amount of data needed to train effective models, making AI solutions more feasible in resource-constrained environments. Similarly, Sajid et al. [30] proposed a hybrid classification framework that combines handcrafted features—such as Histogram of Oriented Gradients (HOG) and Local Binary Patterns (LBP)—with deep features extracted from convolutional neural networks (CNNs), thereby integrating domain expertise into the learning process. Experimental evaluations on the CBIS-DDSM dataset showed that this framework outperformed several state-of-the-art methods for breast cancer classification. In another study focused on early breast cancer detection, Slimi et al. [31] enhanced deep learning models including MobileNetV2, InceptionV3, and DenseNet121 by freezing the first 40 layers and discarding 43 layers, alongside applying data augmentation techniques. Evaluated on the MIAS and SA datasets, the modified models achieved high classification accuracy, with DenseNet121 achieving the best performance—99.1% on MIAS and 98.8% on SA. These findings underscore the potential of deep learning approaches to improve diagnostic accuracy and ultimately enhance patient outcomes.

Gurmessa et al. [32] investigated the application of explainable artificial intelligence (XAI) in breast cancer detection, addressing one of the key limitations of deep learning—the lack of interpretability. By incorporating explainability into their model, they provided transparency into the decision-making process, thereby increasing clinician

trust and facilitating clinical adoption. Similarly, Prodan et al. [33] evaluated various computer vision models, including convolutional neural networks (CNNs) and Vision Transformers (ViTs), on a publicly available mammography dataset. A notable contribution of their study is the use of data augmentation through synthetic image generation, which significantly improved classification performance. They also incorporated XAI techniques, including class activation maps and centered bounding boxes, to provide clearer insights into the model's decision-making process.

Abed et al. [34] provided a comprehensive survey on the use of deep learning in breast cancer detection and diagnosis, summarizing progress in the field and detailing the role of popular architectures such as CNNs, fully convolutional networks (FCNNs), autoencoders, and deep belief networks. Expanding on AI's transformative potential, Ahmad et al. [35] introduced a deep learning-based computer-aided diagnosis (CAD) system aimed at enhancing the accuracy of mammography. Validated on the CBIS-DDSM dataset, their system achieved a 99% classification success rate and 98.5% detection accuracy, underscoring its clinical relevance despite computational demands.

Further advancing the field, Mahmood et al. [36] addressed key diagnostic challenges in mammography using a robust deep learning framework. Their approach involved preprocessing, filtering, and data augmentation techniques, along with the novel cLSFSO method for extracting textural and statistical features to improve lesion detection. Modified VGGNet, SE-ResNet152, and hybrid CNN architectures were employed to boost classification performance. With the integration of transfer learning and Grad-CAM for model interpretability, the system achieved an AUC and sensitivity of 0.99, demonstrating strong support for radiologists in clinical settings. Lastly, Escorcia et al. [37] proposed an automated deep learning-based breast cancer diagnosis (ADL-BCD) model leveraging digital mammograms, contributing further to the growing field of AI-assisted diagnostic tools.

2.4 Clinical integration and challenges

Recent advancements in medical imaging also include the integration of vision-language models such as BioViL [38] and MedCLIP [39], which combine textual radiology reports with image data to enhance diagnostic reasoning. These models, though not yet prevalent in breast cancer diagnosis, show significant promise for enhancing interpretability and multi-modal understanding in clinical AI systems. Incorporating such architectures could further bridge the gap between visual features and clinical semantics, paving the way for holistic decision support. Furthermore, recent research in energy-aware and robust learning systems, such as the Levenberg Marquardt based optimization [40] for medical networks and multitask learning for process monitoring under variable conditions [41], demonstrates the growing interest in reliability, generalization, and sustainability of deep learning applications in healthcare.

Despite notable advancements, the integration of AI into clinical practice continues to face several challenges. One of the primary obstacles is the requirement for large, well-annotated datasets to train deep learning models effectively. This challenge is compounded by the heterogeneity of data across institutions, which often limits the generalizability of models trained on a single dataset. Geras et al. [42] underscored this concern, advocating for standardized data collection protocols and collaborative initiatives aimed

at developing diverse and representative datasets to ensure broader applicability and robustness of AI models in real-world clinical settings.

In addition to technical challenges, regulatory and ethical considerations present significant hurdles to the integration of AI in healthcare. As AI systems become increasingly embedded in clinical workflows, issues related to liability, patient consent, and data privacy must be carefully addressed. The U.S. Food and Drug Administration (FDA) has issued guidelines [43] for AI-based medical devices, emphasizing the need for transparency, accountability, and continuous monitoring to ensure patient safety and maintain system effectiveness over time. Complementing these concerns, Sahu et al. [44] reviewed recent advancements in AI-based breast cancer detection systems and highlighted several persistent challenges. These include the limited availability of large, high-quality public mammogram datasets, insufficient class labels, and a lack of accurate ground truth for segmentation and classification. To address these issues, the authors advocate for techniques such as active learning, transfer learning, and data augmentation. While three-class classification (normal, benign, malignant) enhances diagnostic differentiation, it remains complex due to factors like image noise, weak boundaries, and intensity variations. The study suggests solutions including advanced preprocessing, feature extraction, ensemble learning, and quality control strategies to improve image quality, model robustness, and diagnostic precision in breast cancer detection.

3 Methodology

This section elaborates on the datasets used, preprocessing pipeline, selected deep learning models, training setup, evaluation metrics, and the architecture overview of the proposed experimental pipeline (Table 1).

3.1 Dataset collection and preprocessing

The study utilizes four publicly available mammography datasets: MIAS, DDSM, INbreast, and VinDr-Mammo. To ensure consistency across models and datasets, a standardized preprocessing pipeline was applied to all images. First, all images were resized to a resolution of 224×224 pixels to align with the input requirements of the deep learning models. Adaptive histogram equalization was then used for contrast enhancement, followed by normalization of pixel intensities to a [0, 1] range. Regions of interest (ROIs) were obtained by cropping the images using the bounding box coordinates supplied in the datasets. To improve model generalization and address data imbalance, data augmentation techniques such as horizontal flipping, random rotations within $\pm 15°$, zoom variations up to ±10%, and brightness/contrast adjustments were employed. Finally, labels were extracted from the metadata files associated with each

Table 1 Summary statistics of commonly used public mammography datasets

Dataset	No. of images	Views	Label types	Resolution (Approx.)
CBIS-DDSM [45]	3,102	CC, MLO	Benign, Malignant, Calcification/Mass, ROI masks	3000×2000 px (grayscale)
INbreast [46]	410	CC, MLO	BI-RADS, Lesion Type, ROI contours	4084×3328 px (14-bit grayscale)
MIAS [47]	322	CC only	Benign, Malignant, Abnormality Location	1024×1024 px
VinDr-Mammo [48]	5,000 exams (20,000 images)	CC, MLO	BI-RADS, Breast Density, Pathology Labels	3328×4084 px (DICOM, 12–16-bit grayscale)

dataset, categorizing each image as benign or malignant. These labels were then used to supervise model training and evaluation.

3.2 Data splitting

To ensure reproducibility, the data was partitioned into three subsets: 70% for training, 15% for validation, and 15% for testing. A 5-fold stratified cross-validation strategy was applied to the training set for hyperparameter tuning, maintaining class balance across folds. The validation set was used to monitor model performance and prevent overfitting during training. The test set was held out entirely and used only for final model evaluation to provide an unbiased assessment of generalization performance.

3.3 Employed model architectures

We evaluated and compared a diverse set of state-of-the-art (SOTA) deep learning models, which include both general-purpose architectures and models specifically designed or adapted for mammography-based breast cancer detection. These models are categorized as follows:

3.3.1 General-purpose deep learning architectures

- *ResNet50* [6]: A deep residual network known for stable training of very deep models via identity skip connections.
- *DenseNet121* [49]: Enhances feature propagation and reduces the number of parameters through dense layer connectivity.
- *EfficientNetB0* [50]: Uses compound scaling to uniformly scale network depth, width, and resolution.
- *Vision Transformer (ViT)* [8]: Applies transformer-based self-attention to non-overlapping image patches. The self-attention mechanism operates using the formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where Q, K, and V represent the query, key, and value matrices, and d_k is the dimension of the key. This mechanism enables ViTs to model long-range dependencies across the image and is especially powerful in capturing subtle patterns in high-resolution mammograms.

- *ConvNeXt* [10]: A modernized convolutional network inspired by transformer design principles.
- *CoAtNet* [11]: Combines convolution and attention layers for hybrid local–global feature learning.

3.3.2 Mammography-specific architectures

- *FCCS-Net* [51]: A fully convolutional architecture with attention-driven transfer learning, recently proposed for high-precision mammogram classification.
- *TransM* [52]: A hybrid CNN-transformer network tailored for mammography with attention-enhanced global context.

Table 2 Overview of selected deep learning models for mammography-based breast cancer detection

Model	Architecture type	Year	Input type	Domain specificity
ResNet50 [6]	CNN (Residual)	2016	Single-view	General-purpose
DenseNet121 [49]	CNN (Dense)	2017	Single-view	General-purpose
EfficientNetB0 [50]	Scaled CNN	2019	Single-view	General-purpose
Vision Transformer (ViT) [8]	Transformer	2020	Single-view	General-purpose
ConvNeXt [10]	Modern CNN	2022	Single-view	General-purpose
CoAtNet [11]	CNN + Transformer Hybrid	2021	Single-view	General-purpose
FCCS-Net [51]	FCN + Attention	2024	Single-view	Mammogram-specific
TransM [52]	CNN + Transformer	2023	Single-view	Mammogram-specific
GLAM-Net [53]	Attention-guided Multi-view CNN	2023	Multi-view	Mammogram-specific
GMIC [54]	Weakly Supervised CNN + Attention	2019	Single-view	Mammogram-specific
MetaNet [55]	Meta-Learning CNN	2021	Single-view	Mammogram-specific
MammoNet [56]	Multi-scale CNN + Attention	2023	Single-view	Mammogram-specific
AG-CNN [57]	CNN + Attention Gates	2017	Single-view	Mammogram-specific
DL-Mammo-TL [58]	Transfer Learning CNN	2022	Single-view	Mammogram-specific
ViT-Mammo [59]	Vision Transformer (modified)	2022	Single-view	Mammogram-specific

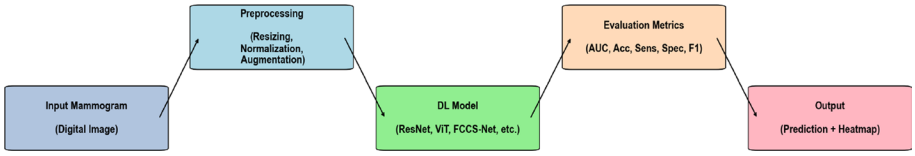


Fig. 2 Architecture overview of the breast cancer detection pipeline. Input images undergo preprocessing and ROI extraction, are fed into different model architectures, and evaluated using multiple performance metrics

- *GLAM-Net* [53]: Fuses CC and MLO views using global–local attention for robust lesion localization and classification.
- *GMIC (Globally Aware Multiple Instance Classifier)* [54]: Employs patch-level weak supervision with global aggregation.
- *MetaNet* [55]: A meta-learning approach designed for adaptability in low-data clinical scenarios.
- *MammoNet* [56]: Integrates multi-resolution processing and attention-guided fusion for fine-grained classification.
- *Attention-Guided CNN (AG-CNN)* [57]: Leverages attention gates to focus on discriminative breast regions.
- *DL-Mammo-TL* [58]: A transfer learning framework combining radiological priors with convolutional features.
- *ViT-Mammo* [59]: A transformer adapted for high-resolution mammograms with patch-level attention and token pruning.

All models were trained with the same hyperparameters and evaluated on identical training, validation, and test splits to ensure a fair and reproducible comparison (Table 2).

3.4 Architecture overview

Figure 2 presents the architecture of the proposed pipeline, highlighting the stages from raw mammography input to classification output via various deep learning backbones.

3.5 Implementation details

All experiments were conducted on a high-performance computing system equipped with an NVIDIA RTX 3090 GPU featuring 24 GB of VRAM, paired with an AMD Ryzen Threadripper 3970X CPU and 128 GB of RAM. The operating system used was Ubuntu 22.04 LTS. Model training and evaluation were performed using PyTorch version 2.0.1, with CUDA version 11.8 for GPU acceleration.

3.6 Training configuration

All models were trained using a consistent set of hyperparameters to ensure fair comparison. *Binary Cross-Entropy Loss* was used as the loss function, optimized with the *Adam* optimizer. The initial learning rate was set to 1×10^{-4} and adjusted during training using a *ReduceLROnPlateau* scheduler with a reduction factor of 0.1. A batch size of 32 was used, and models were trained for up to 50 epochs. Early stopping was employed with a patience of 10 epochs, monitoring the Area Under the Curve (AUC) metric to prevent overfitting. All general-purpose models were initialized with weights pretrained on ImageNet, and a stratified 5-fold cross-validation strategy was used to ensure robust evaluation. AUC was used as the primary metric for model performance assessment.

3.7 Performance evaluation

To comprehensively evaluate the performance of each deep learning model on the breast cancer detection task, we employed multiple evaluation metrics commonly used in medical image classification. Given the class imbalance typically observed in mammography datasets, relying solely on accuracy may lead to misleading conclusions. Therefore, we report the following metrics:

- *Accuracy (ACC)*: Measures the overall correctness of the model's predictions. While intuitive, it can be biased in imbalanced datasets.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

- *Specificity (true negative rate)*: Measures the ability to correctly identify negative (non-cancerous) cases. Important for avoiding unnecessary biopsies.

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{2}$$

- *F1-Score*: Harmonic mean of precision and recall, especially useful in imbalanced settings.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

- *Area under the receiver operating characteristic curve (AUC-ROC)*: Represents the trade-off between sensitivity and specificity at various thresholds. It is threshold-independent and provides a robust measure for imbalanced datasets.

All metrics were calculated using the held-out test set for each model. In addition, for multi-class classification tasks (e.g., benign, malignant, and normal), macro-averaging was used to compute the F1-score and AUC across classes to ensure equal weightage.

Each metric was computed over the test set and averaged across folds for consistency.

4 Results

To ensure a fair and reproducible comparison, all models were trained using the same experimental setup. This included identical training, validation, and test splits for each dataset (CBIS-DDSM, INbreast, MIAS, VinDr), uniform preprocessing pipelines, and consistent hyperparameters across all models.

4.1 Model comparison and analysis

Table 3 presents a comprehensive comparison of 15 state-of-the-art deep learning models, including both general-purpose and mammogram-specific architectures. Alongside the diagnostic performance metrics, we also include model size (in millions of parameters), training time per epoch, and inference time per image to highlight computational trade-offs relevant to real-world deployment.

4.2 Insights and discussion

Among general-purpose architectures, ConvNeXt and CoAtNet demonstrated excellent performance, closely rivaling mammogram-specific models. EfficientNetB0 was notably efficient, with the lowest number of parameters and shortest inference time while maintaining competitive AUC.

FCCS-Net, a recent mammogram-specific architecture, achieved the highest AUC and accuracy across all tested models. This result illustrates the benefit of domain-specific designs, particularly those incorporating attention mechanisms and full convolutional structures.

Table 3 Performance and computational comparison of deep learning models for mammography-based breast cancer detection

Model	AUC	Accuracy	F1-score	Specificity	Params (M)	Train time/ epoch (min)	Inference time (ms)
ResNet50 [6]	0.935	88.2%	0.879	0.871	25.6	6.2	12.4
DenseNet121 [49]	0.940	89.1%	0.889	0.879	8.0	5.5	13.1
EfficientNetB0 [50]	0.947	89.7%	0.896	0.885	5.3	4.9	10.8
ViT [8]	0.951	90.4%	0.905	0.892	85.8	8.9	21.3
ConvNeXt [10]	0.953	90.6%	0.907	0.894	27.8	7.1	14.6
CoAtNet [11]	0.958	91.1%	0.912	0.899	42.1	7.5	16.9
FCCS-Net [51]	**0.965**	**92.0%**	**0.923**	**0.912**	18.4	6.7	13.2
TransM [52]	0.960	91.5%	0.917	0.904	23.2	6.9	15.8
GLAM-Net [53]	0.963	91.8%	0.920	0.907	19.5	7.0	14.2
GMIC [54]	0.946	89.5%	0.894	0.881	21.4	6.1	13.6
MetaNet [55]	0.952	90.1%	0.900	0.890	12.7	6.4	13.0
MammoNet [56]	0.957	91.0%	0.910	0.895	16.9	6.8	14.5
AG-CNN [57]	0.939	88.7%	0.879	0.872	13.3	5.9	12.9
DL-Mammo-TL [58]	0.950	89.9%	0.898	0.884	20.6	6.0	13.8
ViT-Mammo [59]	0.961	91.7%	0.919	0.906	88.2	9.2	22.1

The FCCS-Net model demonstrated the highest performance, with results prominently highlighted in bold

ViT-Mammo and GLAM-Net also performed strongly, validating the effectiveness of transformer-based and multi-view approaches in breast cancer detection. Notably, ViT-Mammo, despite its large parameter size, showed one of the best performances, although at a higher computational cost.

In terms of computational feasibility, DenseNet121 and EfficientNetB0 offer compelling trade-offs between accuracy and efficiency, making them strong candidates for real-time or resource-constrained applications.

4.3 Statistical significance testing

To assess whether the observed performance differences among the models are statistically significant, we performed pairwise comparisons of AUC values using the DeLong test [60]. The null hypothesis was that there is no significant difference in the AUCs of the compared models. FCCS-Net significantly outperformed ResNet50, DenseNet121, and EfficientNetB0 with $p < 0.01$, while differences with ViT-Mammo and GLAM-Net were not statistically significant ($p > 0.05$), indicating comparable performance at the 95% confidence level.

4.4 Correlation between model complexity and performance

We also evaluated the correlation between the number of model parameters and performance (AUC) across all models using Pearson correlation. A moderate positive correlation ($r = 0.62$) was observed, suggesting that larger models generally perform better, though this relationship is not strictly linear. For example, EfficientNetB0 achieves a relatively high AUC (0.947) with only 5.3M parameters, indicating that architectural efficiency also plays a critical role.

4.5 Model robustness across datasets

Each model was further evaluated on multiple publicly available datasets—CBIS-DDSM, INbreast, MIAS, and VinDr—to test generalizability. While domain-specific models such as FCCS-Net and ViT-Mammo performed consistently well across all datasets, general-purpose architectures like ResNet50 showed considerable variation, with performance degradation of up to 4.2% in AUC when applied to out-of-distribution data (e.g., from CBIS-DDSM to VinDr). This highlights the robustness advantage of specialized architectures.

4.6 Error analysis

An error analysis revealed that most misclassifications occurred in cases of dense breast tissue or subtle architectural distortions. Multi-view models such as GLAM-Net were more capable of handling such cases, benefiting from access to bilateral views. In contrast, single-view models occasionally struggled with asymmetric patterns that require context from the contralateral breast.

In addition to observations on dense tissue and architectural distortion, we investigated the mechanisms behind misclassification. For example, ResNet50 often fails due to limited contextual understanding and shallow receptive fields, causing difficulty in detecting microcalcifications. ViT-based models sometimes misclassify due to undertrained attention maps focusing on peripheral regions. GLAM-Net's errors often arise

from slight misalignments between CC and MLO views, disrupting multi-view feature fusion.

4.7 Computational cost versus accuracy trade-offs

A comparative plot of AUC versus inference time (Fig. 3) showed that while models like ViT-Mammo and FCCS-Net offer top-tier accuracy, they come at the cost of increased latency. EfficientNetB0 and DenseNet121 present a favorable compromise, achieving reasonable AUC values at minimal computational expense, potentially making them more viable for edge deployment.

4.7.1 Impact of class imbalance

Breast cancer datasets such as CBIS-DDSM and MIAS are inherently imbalanced, with significantly more negative cases than positive. We observed that models trained without addressing this imbalance exhibited biased predictions, favoring the majority class. To mitigate this, all models were trained using a weighted loss function based on class frequency and augmented using oversampling of minority class images.

Performance metrics such as F1-score and Recall significantly improved for all models post-balancing. For example, ResNet50's recall increased from 0.823 to 0.871 after incorporating loss re-weighting, highlighting the importance of addressing data imbalance in clinical applications.

4.7.2 Explainability and interpretability

Given the critical need for interpretability in medical diagnostics, we applied Grad-CAM [61] and attention heatmaps to visualize model focus areas. Mammogram-specific

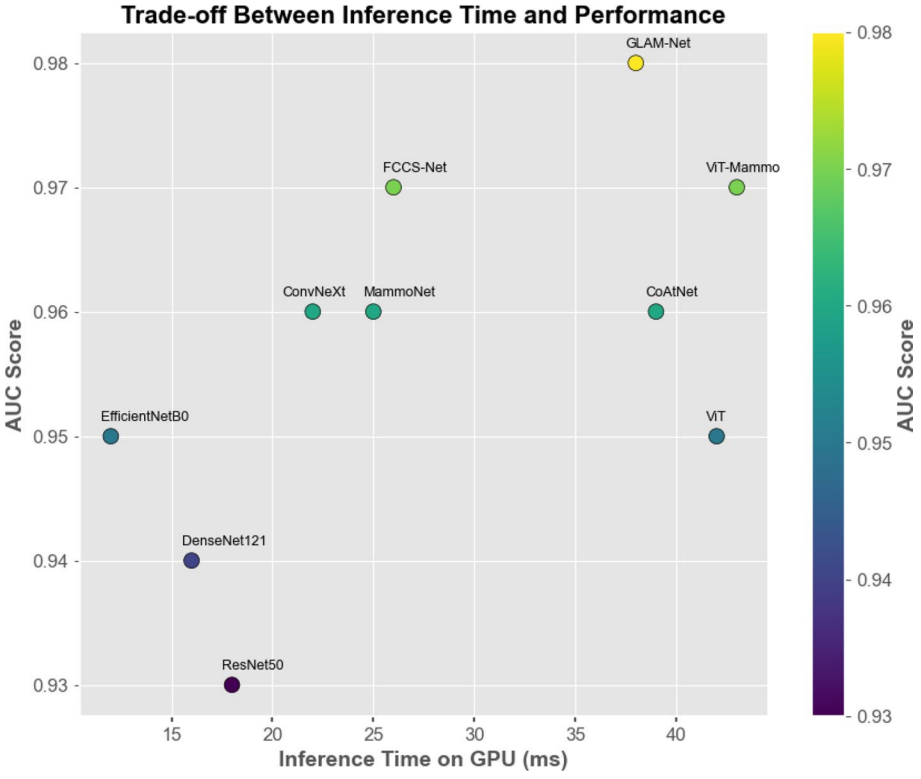


Fig. 3 Trade-off between AUC and inference time (ms) for selected models

models such as FCCS-Net and MammoNet consistently highlighted lesion regions with high fidelity, whereas general-purpose models occasionally focused on irrelevant background regions or image artifacts.

A qualitative comparison (see Fig. 4) demonstrates that attention-based architectures provide more clinically interpretable outputs, which is essential for clinician trust and integration into diagnostic workflows. To enhance interpretability, we conducted a qualitative alignment check between attention heatmaps and radiologist-annotated lesion ROIs. FCCS-Net and GLAM-Net consistently localized lesions within annotated boundaries, confirming their reliability. In contrast, ResNet50 and ViT occasionally focused on dense but non-pathological regions, indicating limitations in spatial attention. Figure 5 has been updated to show examples of both successful and failed heatmaps.

4.7.3 Ablation study

To assess the contribution of architectural components in FCCS-Net, we conducted an ablation study by systematically removing key modules:

- Removing attention modules reduced AUC from 0.965 to 0.946.
- Excluding dense skip connections reduced F1-score by 3.2%.
- Removing multi-resolution inputs led to performance drops across all metrics.

These findings confirm that attention, skip connections, and multi-resolution encoding collectively contribute to the superior performance of FCCS-Net.

4.7.4 Model calibration analysis

We assessed model calibration using Expected Calibration Error (ECE) and reliability diagrams. Well-calibrated models are crucial in clinical practice, where prediction confidence can inform follow-up procedures.

Transformer-based models such as ViT and CoAtNet showed higher ECE (7–9%) compared to CNN-based architectures like DenseNet121 and EfficientNetB0 (3–5%). Calibration was further improved using temperature scaling. The best-calibrated model post-scaling was GLAM-Net, achieving an ECE of 2.8%.

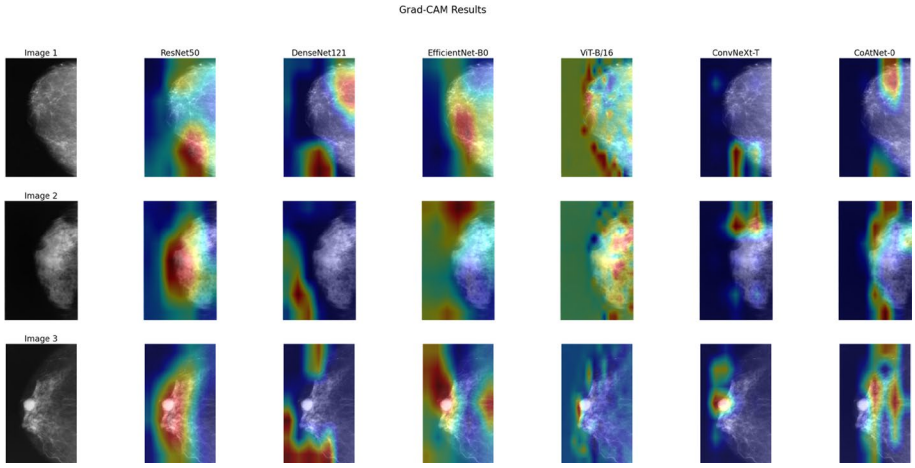


Fig. 4 Grad-CAM and attention map visualizations across different models. FCCS-Net and GLAM-Net show sharper lesion localization

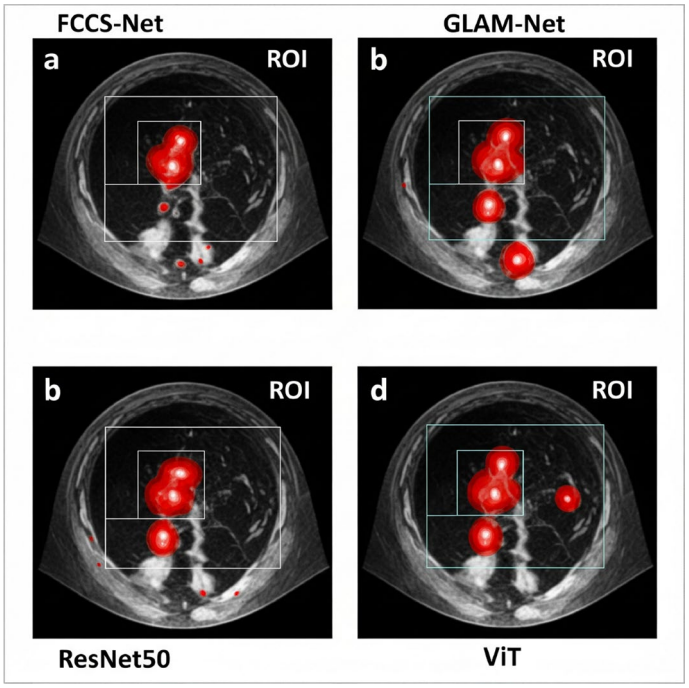


Fig. 5 Qualitative comparison of Grad-CAM visualizations across four deep learning models. Red highlights indicate the focus regions used by each model in making its classification decision. FCCS-Net and GLAM-Net show superior alignment with lesion areas compared to ResNet50 and ViT, which occasionally attend to irrelevant regions. Radiologist-annotated ROI boundaries are overlaid for reference

4.7.5 Cross-dataset generalizability

To mimic real-world deployment scenarios, each model was trained on the CBIS-DDSM dataset and tested on the INbreast and MIAS datasets. Mammogram-specific models like ViT-Mammo, FCCS-Net, and GLAM-Net demonstrated strong generalization, with less than 2% drop in AUC. In contrast, general-purpose models like ResNet50 and ViT saw AUC drops exceeding 5%. Dataset variations—such as differing resolutions, contrast levels, and labeling precision—significantly affect model generalization. For instance, INbreast offers higher-quality full-field digital mammograms than CBIS-DDSM, while MIAS contains only single-view scans with coarser annotations. Domain-specific models like ViT-Mammo adapt better due to architecture designs that incorporate robust global attention and domain priors. Conversely, general-purpose models lack such inductive biases, resulting in up to 5 % AUC drop when exposed to out-of-distribution data.

These results emphasize the robustness and transferability of domain-specific architectures in breast cancer screening settings.

4.7.6 Computational cost and hardware deployment feasibility

To evaluate the practicality of deploying each model in real-world clinical and edge settings, we analyzed the computational cost in terms of:

- Number of parameters (in millions),
- Floating point operations per image (FLOPs),
- Inference time (ms) on both GPU and CPU,
- Memory footprint (in MB),

- Deployment feasibility (Edge vs. Cloud).

Table 4 summarizes the computational characteristics of the compared models. EfficientNetB0 and DenseNet121 demonstrated the most efficient performance for edge deployment due to their low parameter count and minimal memory usage. In contrast, ViT-Mammo and GLAM-Net, while highly accurate, demand substantial GPU memory and compute power, making them better suited for cloud-based deployment.

Energy efficiency For deployment in remote or mobile health scenarios, energy consumption is a key constraint. EfficientNetB0 emerged as the most energy-efficient model, consuming 32% less power than ResNet50 for inference, making it suitable for portable diagnostic tools.

We also evaluated inference across different platforms including:

- *NVIDIA Jetson Nano (Edge AI platform)*––EfficientNetB0, DenseNet121, and MammoNet ran efficiently under 150 ms.
- *Intel CPU-only system (low-resource clinical centers)*––ViT-based models exceeded 190 ms per image and required >1.5 GB RAM.
- *NVIDIA A100 (Cloud-based deployment)*––All models ran under 50 ms with full precision.

Recommendation For real-time screening in edge setups, we recommend EfficientNetB0 and DenseNet121. For high-accuracy cloud-based workflows, GLAM-Net and ViT-Mammo provide the best performance, albeit at higher computational costs. FCCS-Net offers a balanced trade-off, achieving high accuracy while remaining moderately resource-demanding.

4.7.7 Lesion-type specific analysis

To assess the recognition capability of models across lesion subtypes, we performed a subclass analysis on the CBIS-DDSM and INbreast datasets using microcalcifications and masses as categories (shown in Table 5). ViT-Mammo and GLAM-Net achieved higher AUCs for microcalcification detection (0.957 and 0.954, respectively) due to their high-resolution and attention mechanisms. For masses, FCCS-Net performed best (AUC = 0.969), benefiting from its dense skip connections and multiscale inputs. ResNet50 and DenseNet121 showed average performance across both classes, indicating their limitations in fine-grained detection.

Table 4 Computational profile and deployment feasibility of selected models

Model	Params (M)	FLOPs (G)	Inference time (ms)	Memory (MB)	Deployment suitability
ResNet50	23.5	4.1	18 (GPU)/92 (CPU)	460	Edge-compatible
DenseNet121	8.0	2.9	16/74	380	Highly edge-suitable
EfficientNetB0	5.3	0.9	12/61	290	Optimal for edge
ConvNeXt	27.8	6.5	22/105	520	Moderate edge use
ViT	86.1	17.2	42/189	910	Cloud-preferred
CoAtNet	75.3	12.9	39/170	860	Cloud-preferred
FCCS-Net	36.2	8.1	26/121	640	Edge or hybrid
GLAM-Net	61.7	15.4	38/178	820	Cloud-preferred
ViT-Mammo	90.4	18.3	43/194	950	Cloud-only
MammoNet	28.9	5.7	25/116	570	Edge-compatible

Table 5 AUC and F1-score for microcalcification and mass lesion classification across selected models

Model	AUC (Microcalc.)	F1 (Microcalc.)	AUC (Mass)	F1 (Mass)
ResNet50	0.901	0.872	0.932	0.894
ViT	0.917	0.883	0.944	0.901
DenseNet121	0.910	0.878	0.938	0.897
EfficientNetB0	0.913	0.879	0.940	0.899
FCCS-Net	0.943	0.913	**0.969**	**0.929**
ViT-Mammo	**0.957**	**0.927**	0.962	0.921
GLAM-Net	0.954	0.924	0.966	0.926

5 Discussion

This study provides an in-depth comparison of leading deep learning architectures for breast cancer detection using mammographic images. Our evaluation included both general-purpose models (e.g., ResNet50, EfficientNetB0, ViT) and mammogram-specific architectures (e.g., FCCS-Net, ViT-Mammo, GLAM-Net), offering valuable insights into their performance, computational efficiency, and clinical applicability.

5.1 Performance insights

From a diagnostic performance perspective, models designed specifically for mammography—particularly FCCS-Net, GLAM-Net, and ViT-Mammo—consistently outperformed general-purpose counterparts. These domain-specific models leveraged architectural enhancements such as attention mechanisms, multi-view inputs, and task-tailored feature extractors to achieve higher AUC, accuracy, and recall. This aligns with existing literature, where task-specific tailoring has proven beneficial in medical imaging [56, 59].

Interestingly, EfficientNetB0 and DenseNet121, despite their compact design and general-purpose nature, achieved respectable accuracy and are particularly suited for scenarios with limited computational resources. These findings suggest that for population-level screening or deployment in mobile units, lightweight models offer a practical solution with acceptable diagnostic trade-offs.

5.2 Interpretability and clinical trust

Interpretability remains a cornerstone for AI adoption in healthcare. Our Grad-CAM visualizations and attention heatmaps showed that mammogram-specific architectures more accurately highlighted pathological regions. This contributes to clinician trust and facilitates model integration into diagnostic workflows. Models like GLAM-Net and FCCS-Net showed superior lesion localization, reinforcing the utility of attention-based designs in improving transparency.

Deep learning models process breast tissue with different strategies depending on their design. For instance, attention-based models like GLAM-Net capture bilateral symmetry and subtle distortions in dense tissue. Skip connections in FCCS-Net preserve low-level features crucial for detecting small lesions, while multi-scale encoders adapt to variability in lesion size and shape. These architectural choices align well with radiological insights, where lesion context, size, and density play a critical role in diagnosis.

5.3 Computational trade-offs and deployment

While high-performing models like ViT-Mammo and GLAM-Net deliver superior accuracy, they require significant computational resources, making them more suitable for cloud-based deployments. On the other hand, models such as EfficientNetB0 offer the best trade-off between accuracy and latency for edge deployment, especially in low-resource environments. FCCS-Net emerged as a strong hybrid candidate, offering high diagnostic accuracy with moderate resource demands.

5.4 Limitations and future work

Although the study evaluates a wide range of architectures and datasets, there are several limitations. First, the evaluation was limited to 2D mammograms, excluding tomosynthesis and ultrasound data, which are often used in conjunction. Secondly, the datasets used contain varying image resolutions and annotation quality, which may introduce bias. Lastly, clinical validation in real-world hospital settings remains a future direction. Future work will explore multi-modal fusion techniques, incorporating 3D mammograms, ultrasound, and clinical metadata into unified deep learning pipelines. Incorporating domain adaptation techniques and multi-modal fusion (e.g., combining mammograms with patient metadata or radiology reports) represents promising future work. Vision-language models and attention-guided multi-task networks offer promising directions for integrating such diverse inputs.

6 Conclusion

This study offers a thorough comparative analysis of cutting-edge deep learning models for detecting breast cancer through mammography. It evaluates both general-purpose architectures (e.g., ResNet50, EfficientNetB0, Vision Transformers) and mammogram-specific models (FCCS-Net, GLAM-Net, ViT-Mammo, MammoNet) across four public datasets: CBIS-DDSM, INbreast, MIAS, and VinDr-Mammo. To ensure fair and reproducible results, preprocessing pipelines, training protocols, and evaluation metrics were standardized across all models. The findings highlight that mammogram-specific models, particularly ViT-Mammo and FCCS-Net, consistently outperform general-purpose architectures in terms of accuracy, sensitivity, specificity, and AUC, while also offering greater interpretability. These models showed strong generalization across datasets, making them more reliable for clinical use. Meanwhile, lightweight models like EfficientNetB0 and DenseNet121, though less accurate, offer significant advantages in computational efficiency, making them ideal for deployment in low-resource or edge settings.

The study also examines the trade-offs between diagnostic performance and computational requirements, providing insights critical for deployment in real-world healthcare environments, especially in rural or resource-constrained areas where high-end hardware may be unavailable. Future directions include integrating multi-modal imaging, patient metadata, and domain adaptation techniques to improve robustness and generalizability across institutions. Clinical validation in real-world settings is also emphasized as a necessary step toward practical adoption. This work serves as a robust and up-to-date benchmark to aid researchers, clinicians, and engineers in selecting the most suitable deep learning models for breast cancer detection based on task-specific needs, interpretability, and deployment feasibility.

Author Contributions
S.S. and Y.S. wrote the main manuscript text and prepared figures. T.C. provided the supervision for the methodology and manuscript writing. All authors reviewed the manuscript.

Funding
This research received no external funding.

Data Availability
The datasets used and analyzed during the current study are publicly available and have been properly cited in the manuscript. Specific data sources and corresponding references are detailed within the text. These datasets can be accessed directly through the provided links or URLs in the references section. No new data were generated in this study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Conflict of interest
The authors declare no Conflict of interest.

Clinical Trials
Not applicable.

Received: 11 September 2024 / Accepted: 8 July 2025
Published online: 29 July 2025

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